A SYSTEM FOR CRYPTANALYSIS BASED ON PARAMETRIC DENOISING OF A PUBLIC KEY

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Title: A SYSTEM FOR CRYPTANALYSIS BASED ON PARAMETRIC DENOISING OF A PUBLIC KEY

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Abstract

A system and method for cryptanalysis that leverages modular signal analysis to identify the prime factors of a large composite number, such as an RSA public key. The invention treats the public key as a complex signal, and by applying a series of parameterized modular filters, it systematically denoises this signal. These filters exploit the inherent mathematical properties of prime numbers and their distribution to identify "impossible overlaps" or regions where prime factors cannot exist. The system then focuses computational resources on the remaining high-probability regions within the number space, significantly reducing the search space for the prime factors and thus enabling an accelerated factorization. This method represents a novel approach to cryptanalysis, moving beyond brute-force factorization to a more intelligent, signal-driven attack.

Background of the Invention

The security of widely used public-key cryptographic systems, such as RSA, relies on the computational difficulty of factoring large composite numbers into their prime factors. While multiplying two large prime numbers is computationally trivial, reversing this process (factorization) is extremely difficult for current classical computers. Attackers traditionally employ general-purpose factorization algorithms, such as the General Number Field Sieve, which have superexponential time complexity. As the size of cryptographic keys increases, the time required to factor them grows exponentially, making these systems practically secure against current classical computational power. However, the discovery of a method that can significantly reduce the search space for prime factors, or exploit hidden patterns in their distribution, could compromise the security of these systems.

Summary of the Invention

The present invention introduces a novel cryptanalysis system and method based on "parametric denoising" of a public key. The system takes a large composite number (e.g., an RSA public key) as its input, treating it as a complex, noisy signal within a vast numerical space. The core of the invention lies in applying a series of modular filters, each defined by a specific set of parameters (divisors, multi-state mapping rules, and pattern recognition algorithms).

These filters are designed to exploit the mathematical "harmonics" and "impossible overlaps" inherent in the distribution of prime numbers. By systematically applying these filters, the system progressively identifies and eliminates regions within the number space where the prime factors cannot possibly reside. This process effectively "denoises" the original signal, focusing the remaining search space onto high-probability regions. The system then employs targeted search algorithms (e.g., optimized trial division or elliptic curve factorization methods) on these reduced regions, dramatically accelerating the factorization process. This approach transforms cryptanalysis from a brute-force problem into a signal processing and pattern recognition challenge, making previously intractable factorization problems computationally feasible.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a flowchart of the overall cryptanalysis method, showing the steps of public key ingestion, parametric denoising, and targeted factorization.

FIG. 2 is a conceptual diagram illustrating the parametric denoising process, showing how modular filters reduce the search space.

FIG. 3 is a diagram illustrating the identification of "impossible overlaps" within the number space, showing regions where prime factors cannot exist.

Detailed Description of the Invention

Referring to **FIG. 1**, the cryptanalysis method begins with the ingestion of a public key (101), which is a large composite number. This public key is treated as a signal within a vast numerical search space. The system then initiates a parametric denoising process (102). This involves applying a series of modular filters, each characterized by specific divisors and multi-state mapping rules. The output of the denoising process is a set of high-probability regions (103) within the numerical space where the prime factors are likely to reside. Finally, a targeted factorization algorithm (104) is applied to these high-probability regions, leading to the efficient discovery of the prime factors (105) of the public key.

Referring to **FIG. 2**, the parametric denoising process is illustrated. A large numerical search space (201), representing the potential range for prime factors, is shown. Modular filters (202), each with specific parameters (e.g., divisor D1, mapping M1; divisor D2, mapping M2), are applied. Each filter eliminates a portion of the search space (shaded areas in 203) by identifying patterns in the modular signals that are inconsistent with prime number

properties. The iterative application of these filters progressively reduces the search space, leaving only a small set of high-probability regions (204).

Referring to **FIG. 3**, the identification of "impossible overlaps" is conceptualized. The number line (301) is shown with various modular patterns. For example, a "Divisor 3 Pattern" (302) might indicate that certain numbers are always divisible by 3. Similarly, a "Divisor 5 Pattern" (303) indicates divisibility by 5. The regions where these patterns *overlap* in a way that is mathematically impossible for a prime number (e.g., a number is divisible by both 3 and 5, and not 3 or 5 itself) are identified as "Impossible Overlaps" (304). These regions are then excluded from the search space for prime factors, as they cannot contain the required prime numbers.

Claims

What is claimed is:

1. A method for cryptanalysis of a composite number with unknown prime factors, comprising the steps of:
    - receiving a composite number;
    - treating the composite number as a signal within a numerical search space;
    - applying a plurality of parametric modular filters to the numerical search space, wherein each filter is configured to:
        - compute a sequence of remainders by applying a modulo operation to integers within a portion of the numerical search space using a predetermined divisor;
        - convert the sequence of remainders into a multi-state digital signal;
        - analyze the digital signal to identify patterns inconsistent with prime number properties, thereby defining regions of "impossible overlaps"; and
    - progressively reducing the numerical search space by eliminating said regions of "impossible overlaps"; and
    - applying a factorization algorithm to the reduced numerical search space to identify the prime factors of the composite number.
2. The method of claim 1, wherein the composite number is an RSA public key.

3. The method of claim 1, wherein the parametric modular filters are applied iteratively.

4. The method of claim 1, wherein the multi-state digital signal comprises four states, each corresponding to a quartile of the divisor's range.

5. The method of claim 1, wherein the step of analyzing the digital signal includes identifying consecutive repetitions of specific states that are indicative of composite numbers.

6. An apparatus for cryptanalysis of a composite number with unknown prime factors, comprising:

   - a receiver configured to obtain a composite number;
   - a memory for storing the composite number and parameters for modular filters;
   - a processor configured to:
     - apply a plurality of parametric modular filters to a numerical search space associated with the composite number, wherein each filter processes a portion of the search space by performing modulo operations and converting remainders into multi-state digital signals;
     - identify regions of "impossible overlaps" within the search space based on patterns in the multi-state digital signals that are inconsistent with prime number properties;
     - reduce the search space by excluding the identified regions;
     - apply a factorization algorithm to the reduced numerical search space to determine the prime factors of the composite number.

7. The apparatus of claim 6, wherein the processor is a massively parallel processor.

Drawings

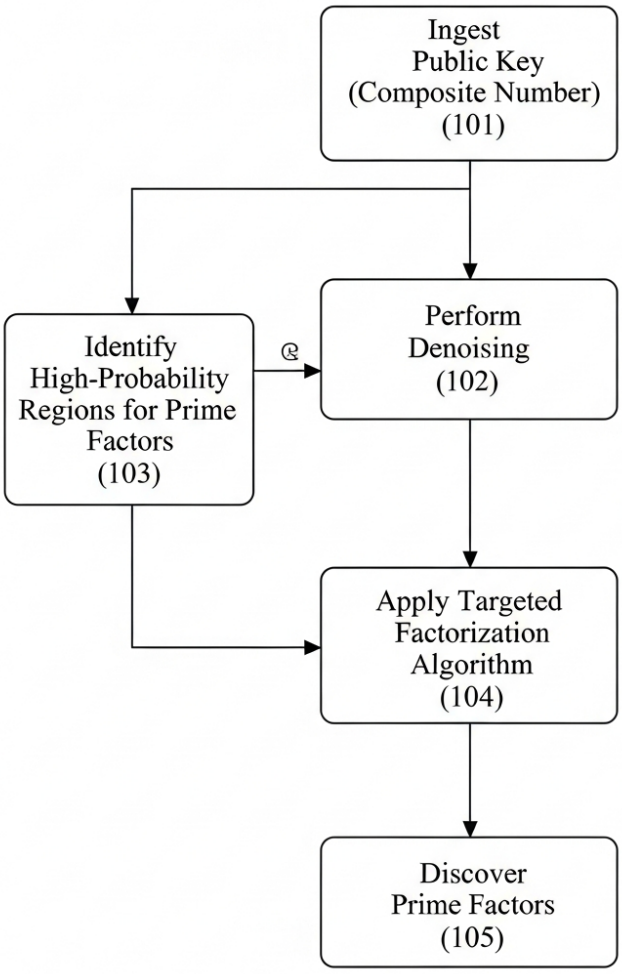


Figure 1: Flowchart of the cryptanalysis method.

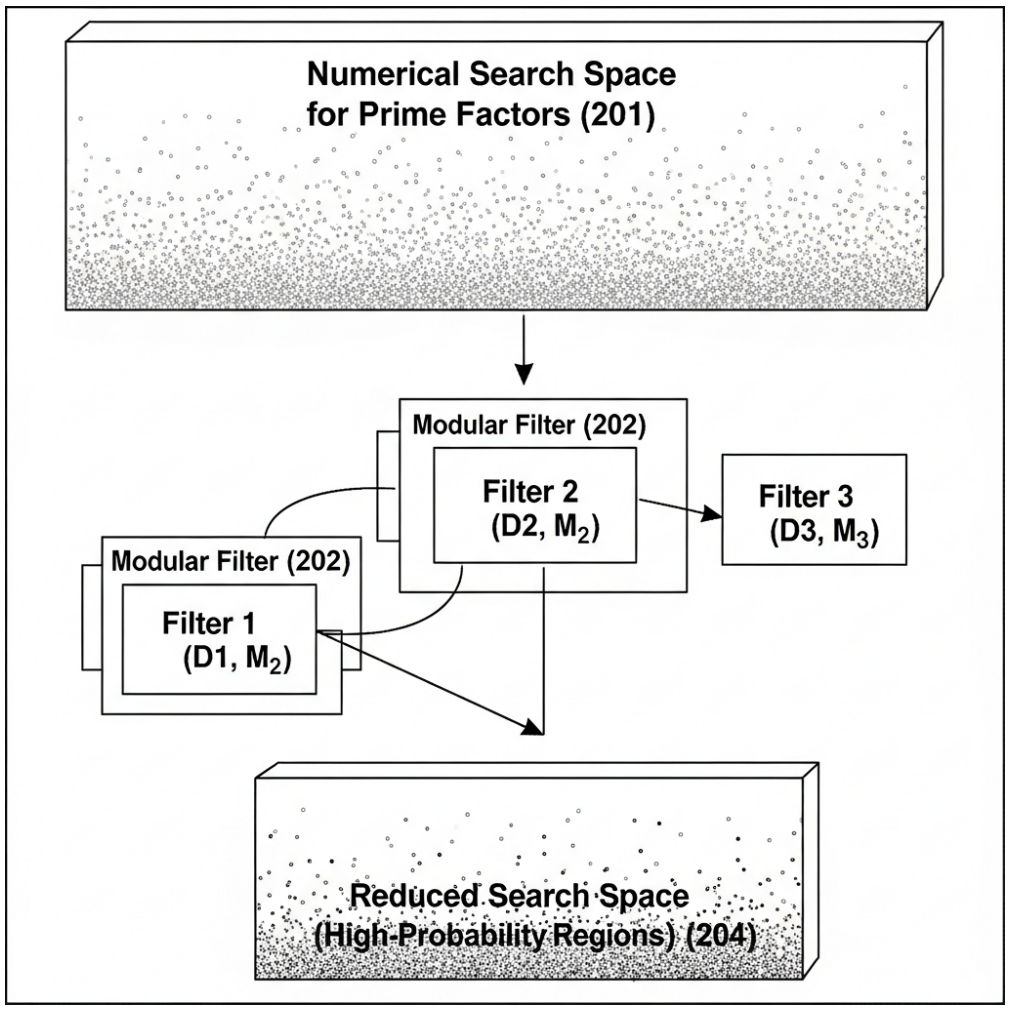


Figure 2: Conceptual diagram of parametric denoising.

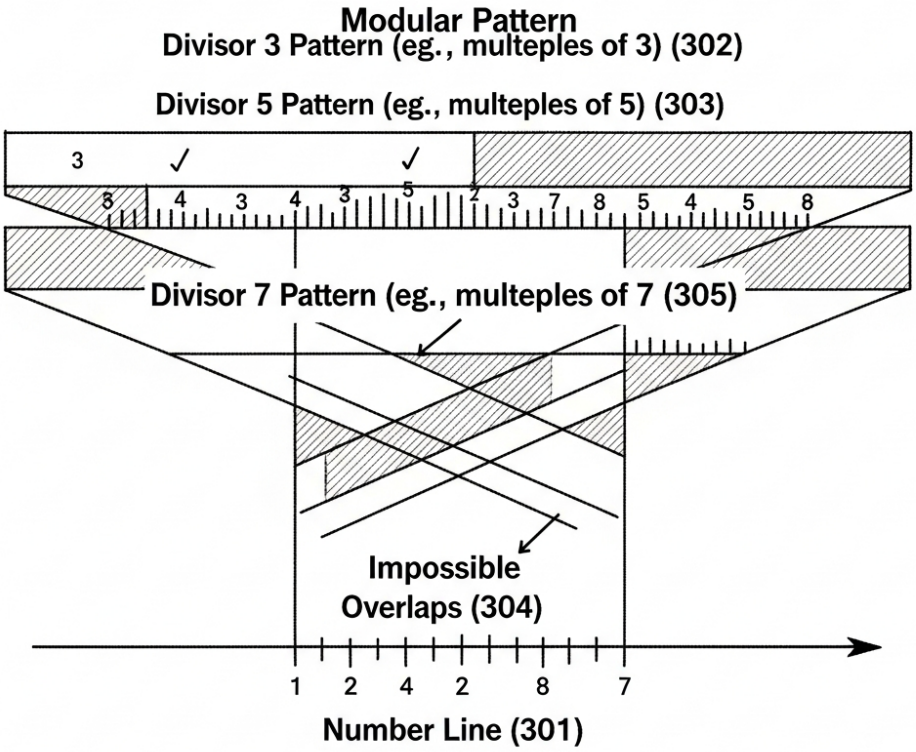


Figure 3: Diagram illustrating impossible overlaps.